

ADVANCED

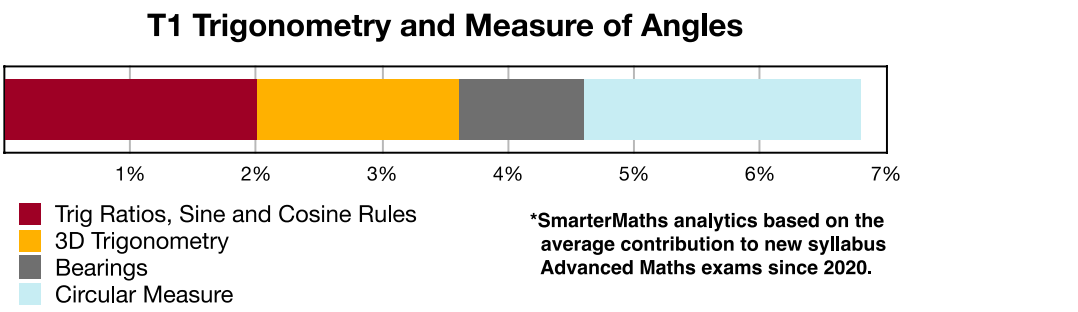
Stage 6

Trigonometry (Adv), T1 Trigonometry and Measure of Angles (Adv)

Circular Measure (Y11)

Teacher: SHS Maths

Exam Equivalent Time: 91.5 minutes (based on allocation of 1.5 minutes per mark)



HISTORICAL CONTRIBUTION

- T1 Trigonometry and Measure of Angles* has contributed a healthy average of 6.8% per Advanced exam since the new syllabus was introduced in 2020.
- This topic has been split into four sub-topics for analysis purposes: 1-*Trig Ratios, Sine and Cosine Rules* (2.0%), 2- *3D Trigonometry* (1.6%), 3-*Bearings* (1.0%) and 4-*Circular Measure* (2.2%).
- This analysis looks at *Circular Measure*.

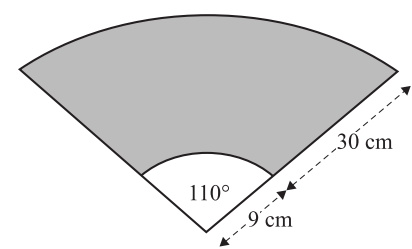
HSC ANALYSIS - What to expect and common pitfalls

- Circular Measure* (2.2%) has been tested in four of the five new syllabus exams. Most recently in 2024 within a cross-topic question with *C3 Maxima/Minima* that appears later in the database.
- Circular Measure* is a fertile area for *Std2/Adv* overlapping content and it will benefit in mark allocations from this (we note *2023 Adv 16* and *2020 Adv 22* appeared in both exams).
- Circular Measure* questions have been well answered in recent years. We do flag this warning: students are encouraged to carefully revise questions in the period 2005-2010 which represent the highest levels of difficulty in this topic area, producing consistent sub-50% mean marks.

Questions

1. Trigonometry, 2ADV T1 SM-Bank 1 MC

A windscreen wiper blade can clean a large area of windscreen glass, as shown by the shaded area in the diagram below.

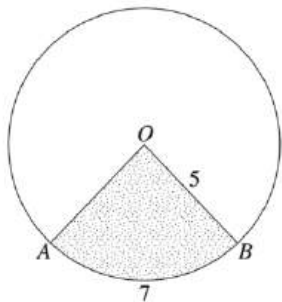


The windscreen wiper blade is 30 cm long and it is attached to a 9 cm long arm.
 The arm and blade move back and forth in a circular arc with an angle of 110° at the centre.
 The area cleaned by this blade, in square centimetres, is closest to

- A. 786
- B. 1382
- C. 2573
- D. 4524

2. Trigonometry, 2ADV T1 2016 HSC 7 MC

The circle centred at O has radius 5. Arc AB has length 7 as shown in the diagram.

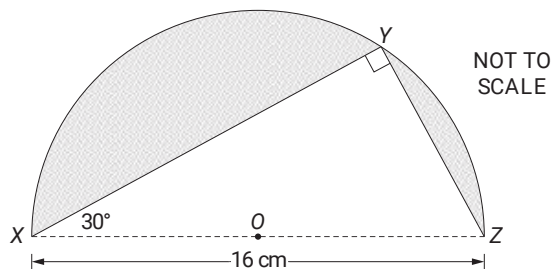


What is the area of the shaded sector OAB ?

- A. $\frac{35}{2}$
- B. $\frac{35}{2}\pi$
- C. $\frac{125}{14}$
- D. $\frac{125}{14}\pi$

3. Trigonometry, 2ADV T1 2021 HSC 12

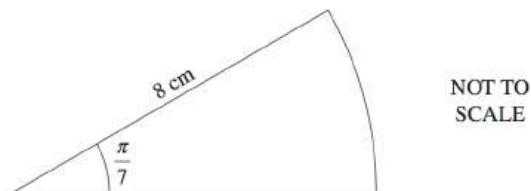
A right-angled triangle XYZ is cut out from a semicircle with centre O . The length of the diameter XZ is 16 cm and $\angle YXZ = 30^\circ$, as shown on the diagram.



- Find the length of XY in centimetres, correct to two decimal places. (2 marks)
- Hence, find the area of the shaded region in square centimetres, correct to one decimal place. (3 marks)

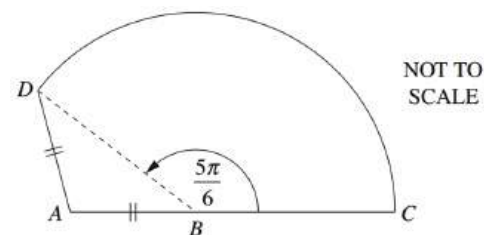
4. Trigonometry, 2ADV T1 2014 HSC 11g

The angle of a sector in a circle of radius 8 cm is $\frac{\pi}{7}$ radians, as shown in the diagram.



Find the exact value of the perimeter of the sector. (2 marks)

5. Trigonometry, 2ADV T1 2006 HSC 4a



In the diagram, $ABCD$ represents a garden. The sector BCD has centre B and $\angle DBC = \frac{5\pi}{6}$

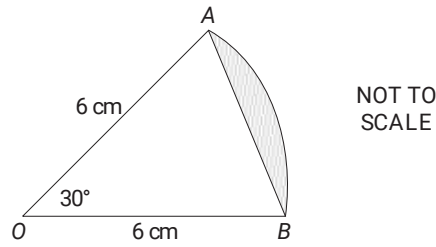
The points A , B and C lie on a straight line and $AB = AD = 3$ metres.

Copy or trace the diagram into your writing booklet.

- Show that $\angle DAB = \frac{2\pi}{3}$. (1 mark)
- Find the length of BD . (2 marks)
- Find the area of the garden $ABCD$. (2 marks)

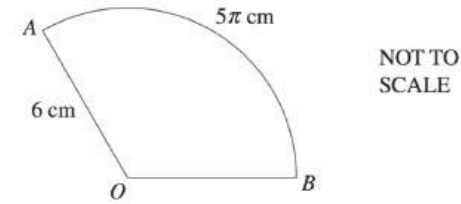
6. Trigonometry, 2ADV T1 2017 HSC 11e

In the diagram, OAB is a sector of the circle with centre O and radius 6 cm, where $\angle AOB = 30^\circ$.



- i. Find the exact value of the area of the triangle OAB . (1 mark)
- ii. Find the exact value of the area of the shaded segment. (1 mark)

7. Trigonometry, 2ADV T1 2004 HSC 4a



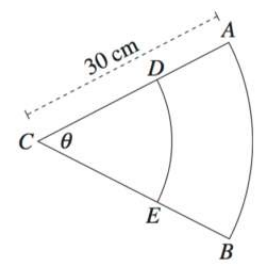
AOB is a sector of a circle, centre O and radius 6 cm.
The length of the arc AB is 5π cm.
Calculate the exact area of the sector AOB . (2 marks)

8. Trigonometry, 2ADV T1 2012 HSC 11f

The area of the sector of a circle with a radius of 6 cm is 50 cm^2 .
Find the length of the arc of the sector. (2 marks)

9. Trigonometry, 2ADV T1 2013 HSC 13c

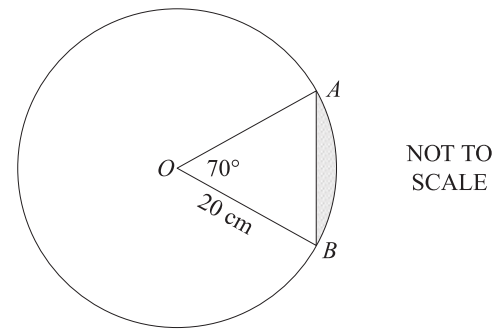
The region ABC is a sector of a circle with radius 30 cm, centred at C . The angle of the sector is θ . The arc DE lies on a circle also centred at C , as shown in the diagram.



The arc DE divides the sector ABC into two regions of equal area.
Find the exact length of the interval CD . (2 marks)

10. Trigonometry, 2ADV T1 2019 HSC 13b

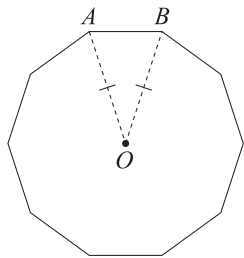
The diagram shows a circle with centre O and radius 20 cm.
The points A and B lie on the circle such that $\angle AOB = 70^\circ$.



Find the perimeter of the shaded segment, giving your answer correct to one decimal place. (3 marks)

11. Trigonometry, 2ADV T1 2020 HSC 22

The diagram shows a regular decagon (ten-sided shape with all sides equal and all interior angles equal). The decagon has centre O .

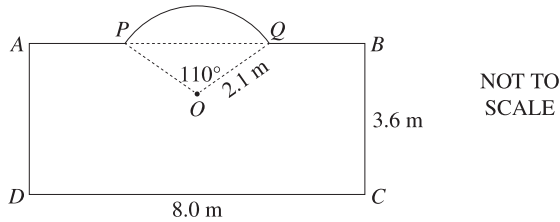


The perimeter of the shape is 80 cm.
By considering triangle OAB , calculate the area of the ten-sided shape. Give your answer in square centimetres correct to one decimal place. **(4 marks)**

12. Trigonometry, 2ADV T1 2023 HSC 16

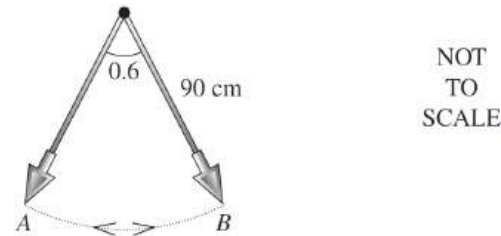
The diagram shows a shape $APQBCD$. The shape consists of a rectangle $ABCD$ with an arc PQ on side AB and with side lengths $BC = 3.6$ m and $CD = 8.0$ m.

The arc PQ is an arc of a circle with centre O and radius 2.1 m and $\angle POQ = 110^\circ$.



What is the perimeter of the shape $APQBCD$? Give your answer correct to one decimal place. **(4 marks)**

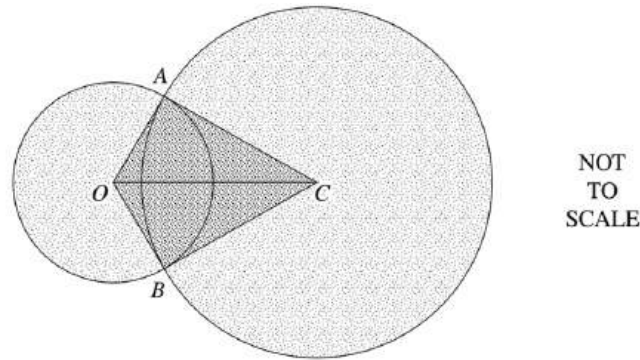
13. Trigonometry, 2ADV T1 2005 HSC 4a



A pendulum is 90 cm long and swings through an angle of 0.6 radians. The extreme positions of the pendulum are indicated by the points A and B in the diagram.

- i. Find the length of the arc AB . **(1 mark)**
- ii. Find the straight-line distance between the extreme positions of the pendulum. **(2 marks)**
- iii. Find the area of the sector swept out by the pendulum. **(1 mark)**

14. Trigonometry, 2ADV T1 2007 HSC 4c



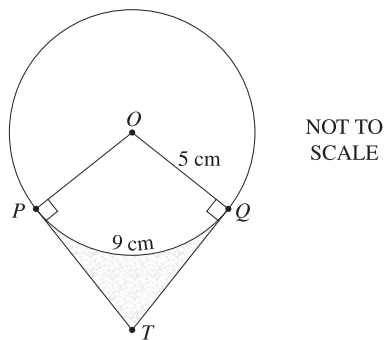
An advertising logo is formed from two circles, which intersect as shown in the diagram. The circles intersect at A and B and have centres at O and C .
The radius of the circle centred at O is 1 metre and the radius of the circle centred at C is $\sqrt{3}$ metres. The length of OC is 2 metres.

- i. Use Pythagoras' theorem to show that $\angle OAC = \frac{\pi}{2}$. **(1 mark)**
- ii. Find $\angle ACO$ and $\angle AOC$. **(2 marks)**
- iii. Find the area of the quadrilateral $AOBC$. **(1 mark)**
- iv. Find the area of the major sector ACB . **(1 mark)**
- v. Find the total area of the logo (the sum of all the shaded areas). **(2 marks)**

15. Trigonometry, 2ADV T1 2010 HSC 6b

The diagram shows a circle with centre O and radius 5 cm.

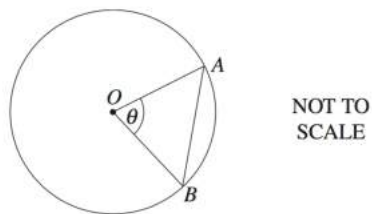
The length of the arc PQ is 9 cm. Lines drawn perpendicular to OP and OQ at P and Q respectively meet at T .



- Find $\angle POQ$ in radians. (1 mark)
- Prove that $\triangle OPT$ is congruent to $\triangle OQT$. (2 marks)
- Find the area of the shaded region. (2 marks)

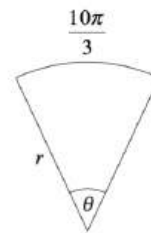
16. Trigonometry, 2ADV T1 2009 HSC 5c

The diagram shows a circle with centre O and radius 2 centimetres. The points A and B lie on the circumference of the circle and $\angle AOB = \theta$.



- There are two possible values of θ for which the area of $\triangle AOB$ is $\sqrt{3}$ square centimetres. One value is $\frac{\pi}{3}$.
Find the other value. (2 marks)
- Suppose that $\theta = \frac{\pi}{3}$.
(1) Find the area of sector AOB (1 mark)
(2) Find the exact length of the perimeter of the minor segment bounded by the chord AB and the arc AB . (2 marks)

17. Trigonometry, 2ADV T1 2008 HSC 7b



The diagram shows a sector with radius r and angle θ where $0 < \theta \leq 2\pi$.

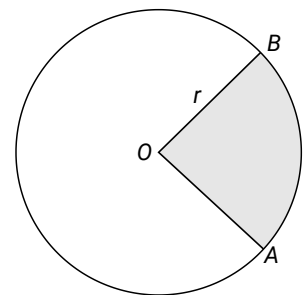
The arc length is $\frac{10\pi}{3}$.

- Show that $r \geq \frac{5}{3}$. (2 marks)
- Calculate the area of the sector when $r = 4$. (2 marks)

18. Trigonometry, 2ADV T1 EQ-Bank 4

The circle below has centre O , radius r and arc length l .

The ratio of the radius to the length of the arc is 2 : 5.



Show that the area of the sector is $A = \frac{5r^2}{4}$. (2 marks)

Worked Solutions

1. Trigonometry, 2ADV T1 SM-Bank 1 MC

$$\begin{aligned}\text{Area cleaned} &= \text{large sector} - \text{small sector} \\ &= \frac{110}{360} \times \pi \times 39^2 - \frac{110}{360} \times \pi \times 9^2 \\ &= 1382.3 \dots \text{ cm}^2 \\ \Rightarrow B\end{aligned}$$

2. Trigonometry, 2ADV T1 2016 HSC 7 MC

$$\begin{aligned}\text{Arc} &= \frac{\theta}{2\pi} \times 2\pi r = r\theta \\ 7 &= 5\theta \\ \therefore \theta &= \frac{7}{5}\end{aligned}$$

$$\begin{aligned}\text{Area of shaded sector} &= \frac{1}{2}\theta r^2 \\ &= \frac{1}{2} \times \frac{7}{5} \times 5^2 \\ &= \frac{35}{2} \text{ u}^2 \\ \Rightarrow A\end{aligned}$$

Worked Solutions

3. Trigonometry, 2ADV T1 2021 HSC 12

$$\begin{aligned}\text{a. } \cos 30^\circ &= \frac{XY}{16} \\ XY &= 16 \cos 30^\circ \\ &= 13.8564 \\ &= 13.86 \text{ cm (2 d.p.)}\end{aligned}$$

$$\begin{aligned}\text{b. Area of semi-circle} &= \frac{1}{2} \times \pi r^2 \\ &= \frac{1}{2} \pi \times 8^2 \\ &= 100.531 \text{ cm}^2\end{aligned}$$

$$\begin{aligned}\text{Area of } \triangle XYZ &= \frac{1}{2} ab \sin C \\ &= \frac{1}{2} \times 16 \times 13.856 \times \sin 30^\circ \\ &= 55.42 \text{ cm}^2\end{aligned}$$

$$\begin{aligned}\therefore \text{Shaded Area} &= 100.531 - 55.42 \\ &= 45.111 \\ &= 45.1 \text{ cm}^2 \text{ (1 d.p.)}\end{aligned}$$

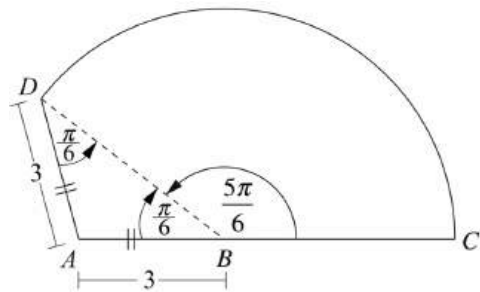
4. Trigonometry, 2ADV T1 2014 HSC 11g

$$\begin{aligned}\text{Arc length} &= \frac{\theta}{2\pi} \times 2\pi r \\ &= \frac{\pi}{7} \times 8 \\ &= \frac{8\pi}{7} \text{ cm}\end{aligned}$$

$$\begin{aligned}\text{Sector perimeter} &= \text{arc} + 2 \times \text{radius} \\ &= \frac{8\pi}{7} + 16 \text{ cm}\end{aligned}$$

5. Trigonometry, 2ADV T1 2006 HSC 4a

i.



$$\text{Show } \angle DAB = \frac{2\pi}{3}$$

$$\begin{aligned}\angle DBA &= \pi - \frac{5\pi}{6} \quad (\pi \text{ radians in straight angle } ABC) \\ &= \frac{\pi}{6} \text{ radians}\end{aligned}$$

$$\therefore \angle BDA = \frac{\pi}{6} \text{ radians} \quad (\text{base angles of isosceles } \triangle ADB)$$

$$\begin{aligned}\therefore \angle DAB &= \pi - \left(\frac{\pi}{6} + \frac{\pi}{6} \right) \quad (\text{angle sum of } \triangle ADB) \\ &= \frac{2\pi}{3} \text{ radians} \dots \text{ as required}\end{aligned}$$

ii. Using the cosine rule:

$$\begin{aligned}BD^2 &= AD^2 + AB^2 - 2 \times AD \times AB \times \cos \frac{2\pi}{3} \\ &= 9 + 9 - (2 \times 3 \times 3 \times -0.5) \\ &= 27 \\ \therefore BD &= \sqrt{27} \\ &= 3\sqrt{3} \text{ m}\end{aligned}$$

$$\begin{aligned}\text{iii. Area of } \triangle ADB &= \frac{1}{2}ab\sin C \\ &= \frac{1}{2} \times 3 \times 3 \times \sin \frac{2\pi}{3} \\ &= \frac{9}{2} \times \frac{\sqrt{3}}{2} \\ &= \frac{9\sqrt{3}}{4} \text{ m}^2\end{aligned}$$

Area of sector BCD

$$= \frac{\frac{5\pi}{6}}{2\pi} \times \pi r^2$$

$$\begin{aligned}&= \frac{5\pi}{12} \times (3\sqrt{3})^2 \\ &= \frac{45\pi}{4} \text{ m}^2\end{aligned}$$

$\therefore$ Area of garden $ABCD$

$$\begin{aligned}&= \frac{9\sqrt{3}}{4} + \frac{45\pi}{4} \\ &= \frac{9\sqrt{3} + 45\pi}{4} \text{ m}^2\end{aligned}$$

6. Trigonometry, 2ADV T1 2017 HSC 11e

$$\begin{aligned}\text{i. Area } \triangle OAB &= \frac{1}{2}ab\sin C \\ &= \frac{1}{2} \times 6^2 \times \sin 30^\circ \\ &= 9 \text{ cm}^2\end{aligned}$$

ii. Area segment = Area sector – Area $\triangle OAB$

$$\begin{aligned}&= \frac{30}{360} \times \pi \times 6^2 - 9 \\ &= 3\pi - 9 \text{ cm}^2\end{aligned}$$

7. Trigonometry, 2ADV T1 2004 HSC 4a

$$\begin{aligned}\text{Arc length} &= \frac{\theta}{2\pi} \times 2\pi r = r\theta \\ 5\pi &= 6\theta \\ \therefore \theta &= \frac{5\pi}{6} \text{ radians}\end{aligned}$$

Area of sector AOB

$$\begin{aligned}&= \frac{\theta}{2\pi} \times \pi r^2 \\ &= \frac{1}{2}r^2\theta \\ &= \frac{1}{2} \times 6^2 \times \frac{5\pi}{6} \\ &= 15\pi \text{ cm}^2\end{aligned}$$

8. Trigonometry, 2ADV T1 2012 HSC 11f

Area of sector, radius 6 cm = 50 cm²

$$\frac{\theta}{2\pi} \times \pi r^2 = 50$$

$$\frac{1}{2} r^2 \theta = 50$$

$$\frac{1}{2} \times 6^2 \times \theta = 50$$

$$\theta = \frac{50}{18} = \frac{25}{9} \text{ radians}$$

$$\begin{aligned} \therefore \text{Length of Arc} &= \frac{\theta}{2\pi} \times 2\pi r \\ &= \theta \times r \\ &= \frac{25}{9} \times 6 \\ &= \frac{50}{3} \text{ cm} \end{aligned}$$

9. Trigonometry, 2ADV T1 2013 HSC 13c

$$\begin{aligned} \text{Area of sector } ABC &= \frac{1}{2} r^2 \theta \\ &= \frac{1}{2} \times 30^2 \times \theta \\ &= 450\theta \end{aligned}$$

Let $CD = x$

$$\text{Area of sector } CDE = \frac{1}{2} x^2 \theta$$

Since DE divides sector ABC in half,

$$\text{Area sector } CDE = \frac{1}{2} \times \text{Area sector } ABC$$

$$\frac{1}{2} x^2 \theta = \frac{1}{2} \times 450\theta$$

$$x^2 = 450$$

$$x = \sqrt{450} \quad (x > 0)$$

$$= 15\sqrt{2} \text{ cm}$$

$\therefore$ The exact length of interval CD is $15\sqrt{2}$ cm.

TIP: Many students find it easier to think of the area of a sector by calculating $\frac{\theta}{2\pi}$ multiplied by the area of a circle rather than remembering a formula.

MARKER'S COMMENT: Simply finding the area of sector ABC achieved half marks in this challenging question! **Show your working.**

10. Trigonometry, 2ADV T1 2019 HSC 13b

$$\begin{aligned} \text{Arc } AB &= \frac{70}{360} \times 2\pi \times 20 \\ &= 24.43\dots \end{aligned}$$

Using cosine rule,

$$\begin{aligned} AB^2 &= 20^2 \times 20^2 - 2 \cdot 20 \cdot 20 \times \cos 70 \\ &= 526.383\dots \\ AB &= 22.94\dots \end{aligned}$$

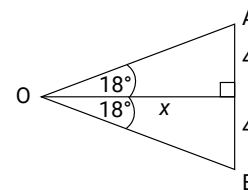
$$\begin{aligned} \therefore \text{Perimeter} &= 24.43 + 22.94\dots \\ &= 47.37 \\ &= 47.4 \text{ cm (1 d.p.)} \end{aligned}$$

11. Trigonometry, 2ADV T1 2020 HSC 22

$$\angle AOB = \frac{360}{10} = 36^\circ$$

$$AB = \frac{80}{10} = 8 \text{ cm}$$

$\triangle AOB$ is made up of 2 identical right-angled triangles



$$\tan 18^\circ = \frac{4}{x}$$

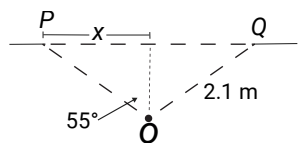
$$x = \frac{4}{\tan 18^\circ}$$

$$\begin{aligned} \therefore \text{Area of decagon} &= 20 \times \frac{1}{2} \times \frac{4}{\tan 18^\circ} \times 4 \\ &= 492.429\dots \\ &= 492.4 \text{ cm}^2 \text{ (to 1 d.p.)} \end{aligned}$$

12. Trigonometry, 2ADV T1 2023 HSC 16

$$\begin{aligned}\text{Arc } PQ &= \frac{110}{360} \times \pi \times 2 \times 2.1 \\ &= 4.03171\dots \text{ m}\end{aligned}$$

Consider $\triangle OPQ$:



$$\begin{aligned}\sin 55^\circ &= \frac{x}{2.1} \\ x &= 2.1 \times \sin 55^\circ \\ &= 1.7202\dots\end{aligned}$$

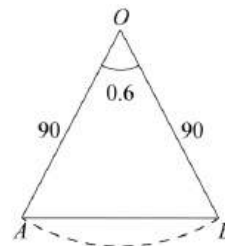
$$PQ = 2x = 3.440 \text{ m}$$

$$\begin{aligned}\therefore \text{Perimeter} &= 8 + (2 \times 3.6) + 4.031 + (8 - 3.440) \\ &= 23.79\dots \text{ m} \\ &= 23.8 \text{ m (to 1 d.p.)}\end{aligned}$$

13. Trigonometry, 2ADV T1 2005 HSC 4a

$$\begin{aligned}\text{i. Arc } AB &= \frac{\theta}{2\pi} \times 2\pi r \\ &= r\theta \\ &= 90 \times 0.6 \\ &= 54 \text{ cm}\end{aligned}$$

ii.



Using the cosine rule

Distance AB in straight line

$$\begin{aligned}AB^2 &= 90^2 + 90^2 - 2 \times 90 \times 90 \times \cos 0.6 \\ &= 2829.563\dots\end{aligned}$$

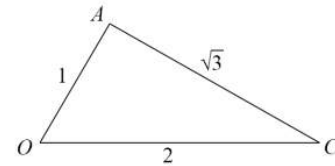
$$\begin{aligned}\therefore AB &= 53.193\dots \\ &= 53.2 \text{ cm (to 1 d.p.)}\end{aligned}$$

iii. Area of Sector

$$\begin{aligned}&= \frac{0.6}{2\pi} \times \pi r^2 \\ &= 0.3 \times 90^2 \\ &= 2430 \text{ cm}^2\end{aligned}$$

14. Trigonometry, 2ADV T1 2007 HSC 4c

i.



In $\triangle AOC$

$$\begin{aligned} AO^2 + AC^2 &= 1^2 + \sqrt{3}^2 \\ &= 1 + 3 \\ &= 4 \\ &= OC^2 \end{aligned}$$

$\therefore \triangle AOC$ is right-angled and $\angle OAC = \frac{\pi}{2}$

ii. $\sin \angle ACO = \frac{1}{2}$

$$\therefore \angle ACO = \frac{\pi}{6}$$

$$\sin \angle AOC = \frac{\sqrt{3}}{2}$$

$$\therefore \angle AOC = \frac{\pi}{3}$$

iii. Area $AOBC$

$$= 2 \times \text{Area } \triangle AOC$$

$$= 2 \times \frac{1}{2} \times b \times h$$

$$= 2 \times \frac{1}{2} \times 1 \times \sqrt{3}$$

$$= \sqrt{3} \text{ m}^2$$

iv. $\angle ACB = \frac{\pi}{6} + \frac{\pi}{6} = \frac{\pi}{3}$

$$\therefore \angle ACB \text{ (reflex)} = 2\pi - \frac{\pi}{3}$$

$$= \frac{5\pi}{3}$$

Area of major sector ACB

$$= \frac{\theta}{2\pi} \times \pi r^2$$

$$= \frac{\frac{5\pi}{3}}{2\pi} \times \pi (\sqrt{3})^2$$

$$= \frac{5\pi}{6} \times 3$$

$$= \frac{5\pi}{2} \text{ m}^2$$

$$\text{v. } \angle AOB = \frac{\pi}{3} + \frac{\pi}{3} = \frac{2\pi}{3}$$

$$\therefore \angle AOB \text{ (reflex)} = 2\pi - \frac{2\pi}{3}$$

$$= \frac{4\pi}{3}$$

Area of major sector AOB

$$= \frac{\frac{4\pi}{3}}{2\pi} \times \pi \times 1^2$$

$$= \frac{2\pi}{3} \text{ m}^2$$

$\therefore$ Total area of the logo

$$= \frac{5\pi}{2} + \frac{2\pi}{3} + \text{Area } AOBC$$

$$= \frac{15\pi + 4\pi}{6} + \sqrt{3}$$

$$= \left(\frac{19\pi + 6\sqrt{3}}{6} \right) \text{ m}^2$$

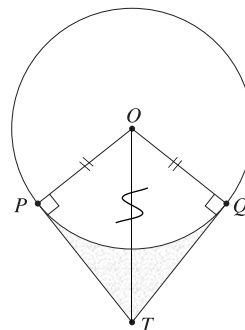
15. Trigonometry, 2ADV T1 2010 HSC 6b

i. Length of Arc $= r\theta$

$$9 = 5 \times \angle POQ$$

$$\therefore \angle POQ = \frac{9}{5} \text{ radians}$$

ii.



Prove $\triangle OPT \cong \triangle OQT$

OT is common

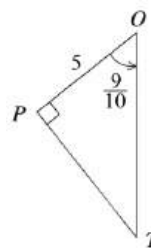
$\angle OPT = \angle OQT = 90^\circ$ (given)

$OP = OQ$ (radii)

$\therefore \triangle OPT \cong \triangle OQT$ (RHS)

MARKER'S COMMENT: Know the difference between the congruency proof of *RHS* and *SAS*. Incorrect identification will lose a mark.

iii.



$$\angle POT = \frac{1}{2} \times \angle POQ \text{ (from part (ii))}$$

$$= \frac{1}{2} \times \frac{9}{5}$$

$$= \frac{9}{10} \text{ radians}$$

$$\tan \angle POT = \frac{PT}{OP}$$

$$\tan \left(\frac{9}{10} \right) = \frac{PT}{5}$$

$$PT = 5 \times \tan \left(\frac{9}{10} \right)$$

$$= 6.3007 \dots$$

♦♦ Mean mark below 30%.

MARKER'S COMMENT: Many students struggled to work in radians. Make sure you understand this concept.

$$= 6.3 \text{ cm (to 1 d.p.)}$$

iv. Shaded Area = Area $OQTP$ – Area Sector OQP

$$\text{Area } OQTP = 2 \times \text{Area } \triangle OPT$$

$$= 2 \times \frac{1}{2} \times OP \times PT$$

$$= 5 \times 6.3007$$

$$\approx 31.503 \dots$$

$$\approx 31.5 \text{ cm}^2$$

$$\text{Area Sector } OQP = \frac{1}{2} r^2 \theta$$

$$= \frac{1}{2} \times 25 \times \frac{9}{5}$$

$$= 22.5 \text{ cm}^2$$

$$\therefore \text{ Shaded Area} = 31.503 - 22.5$$

$$= 9.003 \dots$$

$$= 9.0 \text{ cm}^2 \text{ (to 1 d.p.)}$$

♦ Mean mark 35%.

16. Trigonometry, 2ADV T1 2009 HSC 5c

i. Area $\triangle AOB = \frac{1}{2} ab \sin \theta$

$$= \frac{1}{2} \times 2 \times 2 \times \sin \theta$$

$$= 2 \sin \theta$$

$$2 \sin \theta = \sqrt{3} \text{ (given)}$$

$$\sin \theta = \frac{\sqrt{3}}{2}$$

$$\therefore \theta = \frac{\pi}{3}, \pi - \frac{\pi}{3}$$

$$= \frac{\pi}{3}, \frac{2\pi}{3}$$

$\therefore$ The other value of θ is $\frac{2\pi}{3}$ radians

ii. (1) Area of sector $AOB = \pi r^2 \times \frac{\theta}{2\pi}$

$$= \frac{1}{2} r^2 \theta$$

$$= \frac{1}{2} \times 2^2 \times \frac{\pi}{3}$$

$$= \frac{2\pi}{3} \text{ cm}^2$$

ii. (2) Using the cosine rule:

$$AB^2 = OA^2 + OB^2 - 2 \times OA \times OB \times \cos \theta$$

$$= 2^2 + 2^2 - 2 \times 2 \times 2 \times \cos \left(\frac{\pi}{3} \right)$$

$$= 4 + 4 - 4$$

$$= 4$$

$$\therefore AB = 2$$

$$\text{Arc } AB = 2\pi r \times \frac{\theta}{2\pi}$$

$$= r\theta$$

$$= \frac{2\pi}{3} \text{ cm}$$

$$\therefore \text{ Perimeter} = \left(2 + \frac{2\pi}{3} \right) \text{ cm}$$

17. Trigonometry, 2ADV T1 2008 HSC 7b

i. Show $r \geq \frac{5}{3}$

Arc length = $r\theta$ where $0 < \theta \leq 2\pi$

$$r\theta = \frac{10\pi}{3}$$

$$\therefore \theta = \frac{10\pi}{3r}$$

Using $0 \leq \theta \leq 2\pi$

$$0 \leq \frac{10\pi}{3r} \leq 2\pi$$

$$\frac{10\pi}{3} \leq 2\pi r$$

$$\frac{5}{3} \leq r$$

$$\therefore r \geq \frac{5}{3} \quad \dots \text{as required.}$$

ii. Area = $\frac{1}{2}r^2\theta$

$$= \frac{1}{2} \times 4^2 \times \frac{10\pi}{3 \times 4}$$

$$= \frac{20\pi}{3} \text{ u}^2$$

18. Trigonometry, 2ADV T1 EQ-Bank 4

$$\frac{r}{l} = \frac{2}{5} \Rightarrow l = \frac{5r}{2}$$

$$\text{Area} = \frac{\theta}{360} \times \pi r^2 \dots (1)$$

Arc length (l):

$$\frac{\theta}{360} \times 2\pi r = l$$

$$\frac{\theta}{360} = \frac{l}{2\pi r} = \frac{\frac{5r}{2}}{2\pi r} = \frac{5}{4\pi}$$

Substitute $\frac{\theta}{360} = \frac{5}{4\pi}$ into (1):

$$\text{Area} = \frac{5}{4\pi} \times \pi r^2 = \frac{5r^2}{4} \quad \dots \text{as required}$$